

DECIMALS — ADDITION & SUBTRACTION

Grade 3 Mathematics Lesson

What Are Decimals?

Decimals are numbers with parts smaller than one.

- Look at the number line: 0, 0.1, 0.2, 0.3 ... all the way to 1!
- The dot in the middle is called a **decimal point**.
- Decimals help us show amounts between whole numbers.

Understanding Decimal Place Value

Number	Ones	.	Tenths	Hundredths
3.25	3	.	2	5

Why Do We Use Decimals?

■ Money	■ 25.50
■ Length	1.5 meters
■ Time	9.8 seconds

ADDITION OF DECIMALS

Adding Decimals — The Key Rule

Always **line up the decimal points** before adding! This keeps tenths under tenths and hundredths under hundredths.

Worked Example 1: $1.40 + 2.75$

$$1.40 + 2.75$$

Step 1: $0 + 5 = 5$ (hundredths)

Step 2: $4 + 7 = 11$ (tenths) $\rightarrow$ carry 1

Step 3: $1 + 2 + 1 = 4$ (ones)

✓ **Answer: 4.15**

Worked Example 2: $3.25 + 4.60$

$$3.25 + 4.60$$

Step 1: $5 + 0 = 5$ (hundredths)

Step 2: $2 + 6 = 8$ (tenths)

Step 3: $3 + 4 = 7$ (ones)

✓ **Answer: 7.85**

Worked Example 3: $5.75 + 2.85$

$$5.75 + 2.85$$

Step 1: $5 + 5 = 10 \rightarrow$ write 0, carry 1

Step 2: $7 + 8 + 1 = 16$ (tenths) $\rightarrow$ write 6, carry 1

Step 3: $5 + 2 + 1 = 8$ (ones)

✓ **Answer: 8.60**

Addition Practice Time! ■

Remember: Line up the decimal points first!

1. $2.3 + 1.45 =$ _____

2. $4.6 + 3.2 =$ _____

3. $0.75 + 0.25 =$ _____

4. $6.1 + 2.9 =$ _____

SUBTRACTION OF DECIMALS

Subtracting Decimals — The Key Rule

Just like addition, always **line up the decimal points** before subtracting. Then subtract each column carefully, borrowing when needed.

Worked Example 1: $4.50 - 2.35$

$$4.50 - 2.35$$

Step 1: $0 - 5 \rightarrow$ can't do it! Borrow: $10 - 5 = 5$ (hundredths)

Step 2: $4 - 1$ (borrowed) $- 3 = 0$ (tenths) $\rightarrow$ write 1

Step 3: $4 - 2 = 2$ (ones)

✓ **Answer: 2.15**

Worked Example 2: $6.4 - 3.25$

$$6.4 - 3.25 \rightarrow \text{rewrite as } 6.40 - 3.25$$

Step 1: $0 - 5 \rightarrow$ borrow! $10 - 5 = 5$ (hundredths)

Step 2: $3 - 2 = 1$ (tenths)

Step 3: $6 - 3 = 3$ (ones)

✓ **Answer: 3.15**

Worked Example 3: $7.85 - 4.6$

$$7.85 - 4.6 \rightarrow \text{rewrite as } 7.85 - 4.60$$

Step 1: $5 - 0 = 5$ (hundredths)

Step 2: $8 - 6 = 2$ (tenths)

Step 3: $7 - 4 = 3$ (ones)

✓ **Answer: 3.25**

Subtraction Practice Time! ■

1. $5.5 - 2.25 = \underline{\hspace{2cm}}$

2. $8.3 - 3.1 = \underline{\hspace{2cm}}$

3. $4.75 - 1.5 = \underline{\hspace{2cm}}$

4. $6.9 - 4.4 = \underline{\hspace{2cm}}$

Tips to Remember

1. Always line up the decimal points before adding or subtracting.
2. Add or subtract digits in each place value carefully — hundredths first, then tenths, then ones.
3. Don't forget to bring down the decimal point in your answer.
4. When a column needs borrowing in subtraction, borrow from the column to the left.
5. If one number has fewer decimal places, add a zero as a placeholder (e.g., $6.4 \rightarrow 6.40$).

Summary — What We Learned Today

- Decimals are numbers with parts less than one.
- The dot in the middle is called the **decimal point**.
- Always align decimal points for addition and subtraction.
- Add or subtract digits in each place value carefully.
- Practice makes perfect — keep trying!

You Can Do It! ■

Math is fun when you try your best!